

Coding & Solution

Date	03 July 2026
Team ID	SWTID-2026-5245
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/1010999/Opticrop
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask (backend), Jinja2 (templating)
Key Features Implemented	- ML-based crop recommendation: farmer enters N, P, K, temperature, humidity, pH and rainfall and receives a predicted crop. - Confidence score displayed alongside the recommended crop. - Clean, mobile-responsive input form with client-side range validation (min/max on each field). - Deployed as a serverless web app on Vercel.
Pending / Incomplete Features	- Dedicated crop suitability check (Scenario 2: verifying a chosen crop against given conditions) not yet built as a separate feature. - Researcher/policymaker analytics dashboard (Scenario 3) not yet implemented. - No authentication or user roles. - No live weather API integration (Scenario 1's climate values are entered manually). - No server-side error handling for invalid/out-of-range input.
Setup / Run Instructions	1. Clone the repository and navigate into the project folder. 2. Install dependencies: <code>pip install -r requirements.txt</code> 3. Ensure <code>model/crop_model.pkl</code> and <code>model/label_encoder.pkl</code> are present in the <code>model/</code> folder. 4. To run locally: <code>flask --app api/index.py run</code> (or <code>python api/index.py</code>) 5. To deploy: install the Vercel CLI, then run <code>vercel --prod</code> (vercel.json is already configured to route requests to <code>api/index.py</code>).

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

OptiCrop currently implements the core ML-based crop recommendation flow end-to-end (Scenario 1) and is live on Vercel. The suitability-check and researcher-analytics scenarios, along with input validation and error handling, are identified as next steps before a full production release.